Weil positivity in degree 2:

eight elliptic curves on the prime side — the identity, the rank, the depth, the quorum

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Abstract

The windowed Weil form Q_L of the repository — archimedean part plus prime towers on the window $[-L/2, L/2]$, $L = \log \mu$ — is built for the L-functions of eight elliptic curves (11a1, 19a1, 32a1, 37a1, 43a1, 53a1, 61a1, 67a1) from their a_p alone, and judged against the Gram matrix of their zeros harvested to height 320. After three convention errors and two implementation errors, found by that judge and by an independent frequency-domain evaluation of the archimedean term, the two sides agree to the zero tail (Frobenius 1.4–1.8%) and the smallest eigenvalue agrees to 1–6% on all eight curves and at three windows. Along the way: the prime side reads the analytic rank (the central zero of the four rank-1 curves is required, once, on the constant mode, and refused by the rank-0 curves); the depth s of the well decreases with the conductor ($0.91 \to 0.46$) and with the rank (0.30, 0.25), is desert-dominated (the sub-Nyquist gap term is small because the zeros are twice as dense), and is predicted to $\pm 20\%$ by the desert law with ζ's coefficients; the quorum — every voting prime necessary — holds completely once the well is deep enough ($\ell \gtrsim 10$) and completes progressively before, heaviest towers last; and the two quorum laws of degree 1 hold with the degree as a factor, $-\ln \delta_p \approx 0.19\, d\, s\, p \log p$ and $-\ln \kappa_p \approx 0.11\, d\, s\, (p \log p)^2/(\mu \log \mu)$. Everything is measurement, each claim with its judge in the repository; nothing here bears on RH or on BSD.

1 The form and its judge

For a weight-2 newform f of level N, $\Lambda(s, f) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(s, f)$, critical line $\Re s = 1$, and $\Gamma_{\mathbb{C}}(s) = \Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$. In the analytic normalization the windowed form is

$$Q_L(f) = \sum_{a\in\{0,1\}} \Big[\tfrac{F_0}{2}\big(\tfrac{1}{2}\log N - \log\pi - \gamma - \log(1-e^{-2L})\big) + \int_0^L \frac{F_0 e^{-2y} - e^{-2s_a y}\Theta(y)}{1-e^{-2y}}\,dy\Big] - \sum_{n=p^k\le\mu} \frac{\Lambda_f(n)}{n}\,\Theta(\log n),$$

with $s_0 = \frac{1}{2}$, $s_1 = 1$ (the real parts of $(1+it)/2$ and $(2+it)/2$), $\Lambda_f(p^k) = (\alpha_p^k + \beta_p^k)\log p$ ($\alpha_p + \beta_p = a_p$, $\alpha_p\beta_p = p$; $\alpha_p\beta_p = 0$ for $p \mid N$), Θ the autocorrelation of the test function, $F_0 = \Theta(0)$. Each term was checked: the two panels equal the $\Gamma_{\mathbb{C}}$ term by duplication, $\psi(1+it) = \frac{1}{2}\psi(\frac{1}{2} + \frac{it}{2}) + \frac{1}{2}\psi(1 + \frac{it}{2}) + \log 2$; the conductor contributes $\frac{1}{2}\log N$ once; the $\log(1 - e^{-2L})$ is the tail of the Frullani integral beyond $y = L$, where Θ vanishes but $F_0 e^{-2y}/(1 - e^{-2y})$ does not. The judge is the zero Gram $Q_z = \sum_{\gamma>0} 2\,\hat{c}(\gamma)\hat{c}(\gamma)^\top$ over the harvested zeros ($\hat{c}$ the transforms of the basis; zeros by sign change of the real completed Λ, one-sided Weyl count within 1% to $T = 320$), which under GRH equals Q_L minus the tail beyond 320.

Five errors were found against this judge before agreement: Γ arguments taken at centre $\frac{1}{2}$ instead of 1; $a_{p^k}\log p$ in place of $\Lambda_f(p^k)$; the conductor constant doubled (this alone made a false "positive" form whose bottom mode was the constant function, with the $(0, 0)$ entry $11\times$ the truth); a filter on $a_n = 0$ that dropped the $n = 8$ tower of 11a1 ($a_8 = 0$ but $\Lambda_f(8) = 4\log 2$), located by a scan over lags; and the Frullani tail removed as "ad hoc", located by the frequency-domain evaluation. The a_p come from point counting on Weierstrass models; a wrong model would show at once.

2 The identity and the rank

curve	rank	Frobenius, no central zero	Frobenius, central zero once	$\lambda_{\min}$ prime / zeros
11a1	0	**0.018**	0.123	5.39e-6/5.11e-6 = 1.06
19a1	0	**0.017**	—	2.89e-4/2.78e-4 = 1.04
32a1 (CM)	0	**0.016**	—	8.20e-3/7.96e-3 = 1.03
67a1	0	**0.014**	—	4.35e-2/4.26e-2 = 1.02
37a1	1	0.101	**0.014**	0.396/0.388 = 1.02
43a1	1	0.098	**0.016**	0.867/0.859 = 1.01
53a1	1	0.095	**0.015**	0.869/0.862 = 1.01
61a1	1	0.094	**0.014**	0.958/0.951 = 1.01

The diagonal ratios are all ≥ 1 (the excess is the zero tail; 3.3% estimated on $(0,0)$ for 11a1, 4.3% measured); the residual Frobenius 1.4–1.8% is entirely the tail. The four rank-1 curves fail identically without the central zero — $(0,0)$ seven to twelve times too large — and agree once $\hat{c}(0)\hat{c}(0)^\top$ is added *once* (the central zero is its own conjugate; $\hat{c}(0) = (\sqrt{L}, 0, \dots)$); 11a1 refuses it. The prime-side form, from a_p and Γ alone, reads whether $L(1,E) = 0$. This is the analytic rank, not the Mordell–Weil rank.

3 Depth

Windows $\mu = 11 \to 22$, prime side / zero side ($N = 17/29$):

curve	γ_1	$\lambda(11)$ pr/z	$\lambda(22)$ pr/z	$\hat{s}$ pr/z	N_{eff}
11a1	6.36	5.39e-6 / 5.11e-6	2.48e-10 / 2.22e-10	**0.908**/**0.913**	2.11
19a1	5.04	2.89e-4 / 2.78e-4	3.37e-7 / 3.16e-7	0.614/0.617	1.93
32a1	3.68	8.20e-3 / 7.96e-3	4.96e-5 / 4.73e-5	0.464/0.466	1.69
67a1	3.19	4.35e-2 / 4.26e-2	2.86e-4 / 2.75e-4	0.457/0.458	1.52

The identity holds to 0.5% on the secant. The depth decreases with the conductor and is far below a Dirichlet character of the same γ_1 (χ_5: $\gamma_1 = 6.65$, $s = 2.41$): the zero density is doubled, so the sub-Nyquist gaps that deepen the well in degree 1 are rare here. At a common cutoff $T_0 = 320$ the gap term is 0.2–0.8 of the desert term (against 4–7 in degree 1), the gap sums are small and convergent ($G(22) = 0.5$–2.9 against 20–60), and ζ's coefficients $(a,b) = (1.71, 0.97)$ predict the four depths to $1.07, 0.81, 1.05, 0.92$ — better than the Dirichlet characters at the same cutoff, because the part of the formula that fails there (the gaps) weighs little. The rank-1 curves dig their well orthogonally to the constant mode (weight on η_0 from 0.015 to 0.0000 at $\mu \geq 22$) and are shallower: $s \approx 0.30$ (37a1), 0.25 (43a1).

4 Quorum

Removing all powers of a prime and reading the sign of $\lambda_{\min}$:

curve, window	ℓ	result
11a1, $\mu = 22$	22	every voting prime necessary, whatever the sign of a_p ($a_5 = +1$: -0.22; $a_{13} = +4$: -0.028)
19a1, $\mu = 22$	15	every voting prime necessary; 2 ($a_2 = 0$) necessary through 4 ($\Lambda_f(4) = -4\log 2$)
32a1 (CM), $\mu = 38$	15.5	every voter necessary; 3 ($a_3 = 0$) through 9: -0.63; non-voter 7 invisible to the digit; edge
67a1, $\mu = 22$	8	partial: $2, 5, 13, 17, 19$ dispensable ($+1.02, +0.21, \dots$); $3, 7, 11$ necessary
67a1, $\mu = 38$	10.7	complete: $2, 5, 13$ now $-0.22, -0.60, -0.063$
37a1, $\mu = 22$	4.2	partial: $2, 3, 5$ dispensable (all $a_p < 0$); $7, 11, 13$ necessary
37a1, $\mu = 38$	9.0	$2, 5$ necessary ($-0.33, -0.35$); 3 still dispensable ($+0.37$, falling from $+1.03$)

The quorum is a full-window property: complete once $\ell \gtrsim 10$, progressive before, with the heaviest towers ($|\Lambda_f(p)|/p$ largest) locked last. The switch for 67a1 between $\mu = 22$ and 38 coincides with the recruitment of $23, 29, 31, 37$; nothing moved between 11 and 22 with the same voters. Two predictions died on the way (the "sign of a_p" pattern; a threshold at $\mu = 22$) and one is pre-registered: 37a1 at $\mu = 62$, removal of 3 negative.

5 The quorum laws

On the bottom vector v, $\delta_p = v^\top T_p v$ and $\kappa_p = \|T_p v - \delta_p v\|$, in the prime weight $w = p \log p$ and the window weight $W = \mu \log \mu$. With the degree-1 laws as written ($0.19\, s\, w$) the degree-2 silence comes out 1.8–2× too large; with $s \to d\, s$, $d = 2$:

curve	s	$-\ln \delta_p/w$ $(p = 11, 13, 17)$	$0.19 \cdot 2s$	$-\ln \kappa_p/w^2$ $(p = 17)$	$0.11 \cdot 2s/W$
11a1	0.91	0.37, 0.31, 0.33	0.345	0.00313	0.00294
19a1	0.61	0.23, 0.22, 0.23	0.233	0.00224	0.00199
67a1	0.46	0.12, 0.14, 0.14	0.174	0.00147	0.00148

$-\ln \delta_p \approx 0.19\, d\, s\, p \log p$ and $-\ln \kappa_p \approx 0.11\, d\, s\, (p \log p)^2/(\mu \log \mu)$: the two laws pass with the degree as a factor, to ±10% (11a1, 19a1) and −20% on the shallowest well. What "d" measures — two $\Gamma_{\mathbb{R}}$ factors, doubled zero density, degree of the local Euler polynomial — cannot be told apart between $d = 1$ and 2; degree 3 will.

Status and reproducibility

`code/scan_q_gl2.py` (`GL2_FIX=1`), `code/gl2_curves.py` (models and point counting), zero lists `code/zeros_*a1_weyl.pkl`; `report/scan_q_gl2-review.md` records the five errors; ten recomputing tests (`tests/test_gl2_*.py`). Eight curves, two ranks, three windows: measurement throughout, each claim with its judge. Two pre-registrations died (36th execution of the repository, in magnitude only), one survives (the $d\, s$ laws), two are open on the server (67a1 at $\mu = 74$, 37a1 at $\mu = 62$). The one thing that would have been a result about RH is not here: positivity of Q_L for finite N on curves whose zeros are verified.